spectral lines of Brackett series which lies in infra-red region (middle IR),

$$\bar{v} = \frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{m^2}\right)$$

(e) Pfund series

For $n = 5$ and $m = 6,7,8.....$ in equation (9.18), the wave numbers or wavelength of spectral lines of Pfund series which lies in infra-red region (far IR),

$$\bar{v} = \frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{m^2}\right)$$

Different spectral series are listed in Table 9.2.

Table 9.2

n	m	Series Name	Region
1	2,3,4.....	Lyman	Ultraviolet
2	3,4,5.....	Balmer	Visible
3	4,5,6.....	Paschen	Infrared
4	5,6,7.....	Brackett	Infrared
5	6,7,8.....	Pfund	Infrared

Limitations of Bohr atom model

The following are the drawbacks of Bohr atom model

(a) Bohr atom model is valid only for hydrogen atom or hydrogen like-atoms but not for complex atoms.

(b) When the spectral lines are closely examined, individual lines of hydrogen spectrum are accompanied by a number of faint lines. This is called **fine structure**. This cannot be explained by Bohr atom model.

(c) Bohr atom model fails to explain the intensity variations in the spectral lines.

(d) The distribution of electrons in various levels cannot be completely explained by Bohr atom model.

9.4 NUCLEI

Introduction

In the previous section, we have discussed about various preliminary atom models, Rutherford's alpha particle scattering experiment and Bohr atom model. These played a vital role to understand the structure of the atom and the nucleus. In this section, the structure of the nuclei and their properties, classifications are discussed.

9.4.1 Composition of nucleus

Atoms have a nucleus surrounded by electrons. The nucleus contains protons and neutrons. The neutrons are electrically neutral ($q = 0$) and the protons have positive charge ($q = +e$) equal in magnitude to the charge of the electron ($q = -e$). **The number of protons in the nucleus is called the atomic number** and it is denoted by Z**.** The number of neutrons in the nucleus is called neutron number **(*N*). The total number of neutrons and protons in the nucleus is called the mass number** and it is denoted by A. Hence, $A = Z+N$**.**

The two constituents of nucleus namely neutrons and protons, are collectively called as nucleons. The mass of a proton is 1.6726×10^{-27} kg which is roughly 1836 times the mass of the electron. The mass of a neutron is slightly greater than the mass of the proton and it is equal to 1.6749×10^{-27} kg.

To specify the nucleus of any element, we use the following general notation

$${}^{A}_{Z}X$$

where X is the chemical symbol of the element, A is the mass number and Z is the atomic number. For example, the

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